

THASHMIKA RATHNAYAKE

Software Engineer Intern

[LinkedIn](#) | [GitHub](#) | Dehiwala, Colombo | +94 77 647 4733 | work.thashmika@gmail.com | [Portfolio](#)

Undergraduate - Computer Science

PROFESSIONAL SUMMARY

A result-driven computer science undergraduate in the third year and a passionate programmer who has a good understanding of software performance and great problem solver, possessing a set of good soft skills, seeking a career opportunity as a Software Engineer, focused on enhancing current technical knowledge and using existing knowledge and skills to achieve corporate goals.

TECHNICAL SKILLS

Java | HTML5/CSS3 | JavaScript | React | TypeScript | SQL (MySQL, PostgreSQL) | MongoDB | Git & GitHub | Rest API | Docker | Python

PROJECTS EXPERIENCES

MediGO: Real-time Ambulance Dispatch & Fleet Management Platform

Full-stack Project **Technologies:** TypeScript (React/NestJS), Google Maps API, Firebase (RTDB/Transactions), Flutter

- **The Mission:** Built a high-performance coordination ecosystem to solve emergency response delays and "information silos" between hospitals and ambulance drivers during critical patient transfers.
- **Iterative Agile Development:** Adopted an Agile methodology to rapidly prototype and evolve the platform based on stakeholder feedback, ensuring the "Atomic Mission Dispatch" met critical safety requirements within every sprint.
- **CI/CD Integrity:** Delivered through iterative sprints with a focus on high code quality, implemented a robust CI/CD pipeline using **GitHub Actions** to automate builds and unit testing across Web, Backend, and Mobile services.
- **Concurrent Data Integrity:** Developed "fail-safe" dispatch logic using **NestJS** and **Firebase Transactions**, ensuring atomicity and 100% data consistency during mission-critical real-time assignments.
- **Real-time Resource Optimization:** Integrated a "Pre-check" validation system for hospital resources (ICU/Ventilators), reducing ER redirection rates and improving patient-to-treatment time.
- **Data-Driven Analytics:** Built an automated mission-logging system that uses the Google Routes API (v2) to calculate actual trip distance and estimate fuel consumption (L/km), generating instant "digital trip sheets."
- **Security & Privacy:** Implemented AES-256 encryption for patient records and Role-Based Access Control (RBAC) to ensure all sensitive data remains HIPAA-compliant and secure.

<https://github.com/SDGP-03/MediGo.git>

Mini Service Request Board : Service request marketplace

Full-stack Project **Technologies:** JavaScript (Next.js / Express.js), MongoDB Atlas, Mongoose, Tailwind CSS, Vercel, Railway

- **Overview:** Built a production-ready service request marketplace, enabling homeowners to post home service requests and tradespeople to browse, filter, and manage jobs in real time.
- **RESTful API Design:** Architected and implemented a clean five-endpoint REST API using Node.js and Express.js, following industry-standard HTTP semantics (GET, POST, PATCH, DELETE) with proper status codes and a global error-handling middleware for consistent, secure error responses.

- **Data Modelling & Validation:** Designed a Mongoose schema with multi-layer input validation - enforcing required fields, enum constraints, and regex-based email format checks at the database layer, paired with client-side validation on the frontend to prevent unnecessary API calls.

<https://github.com/realthash/Mini-Service-Request-Board.git>

EDUCATION

University of Westminster, UK | Informatics Institute of Technology (IIT), Sri Lanka 2024 - present
Bachelor of Science (B.Sc.) in Computer Science

Nalanda College - Colombo 10 2014 - 2022

- A/L examination physics(C), Chemistry(S)
- O/L examination 6As and 3Bs

EXPERIENCES

Volunteer Event Coordinator | Rotaract Club of IIT | May 2025

- Contributed to the successful execution of "Dil se," a cultural event for the university community.
- Managed on-site logistics for a segment, serving 50+ participants and ensuring smooth operations.
- Collaborated in a team of 10+ volunteers, demonstrating reliability and effective communication.

EXTRA CARRICULAR ACTIVITIES AND ACHIEVEMENTS

- **Handball (U-19):** Secured **All-Island Runners-Up** position (2018).
- **Handball (U-20):** Achieved **Fourth Place** at the national level (2017).
- **Hockey (U-16):** Crowned **Gold Star Champions** (2016).
- **Handball (U-16):** Attained **All-Island Runners-Up** status (2016).

SKILLS

- **Soft skills** | Problem-Solving | Team Collaboration | Leadership | Brainstorming
- **Languages** | English | Sinhala

CERTIFICATES

- Hult prize Qualifier - 2026
- Python Essential Training – LinkedIn (2025)
- Python Data Analysis (2020) – LinkedIn (2025)
- Python GUI Development with Tkinter – LinkedIn (2025)

REFEREE

Omay Wickramasinghe BSc in Computer Science, UCSC
Senior Software Engineer | DirectFN
Colombo, Sri Lanka
Email: theruni.omaya@directfn.com
Phone: +94 767504104

Dewuruge Nuvin BSc in Computer Science, UCSC
Senior Software Engineer | EY GDS
Colombo, Sri Lanka
Email: nuvin.kalpadeep@gds.ey.com
Phone: +94 765505633